

Unit 2 : Relations and Functions.

• What is Relation and Function ?

Relation and functions are the ways of mapping the establishing link between two entities in math. They are used to establish mathematical relations between two terms.

• Difference between Relation and Function -

Relation	Function
• Relation is defined a non-empty set A to non-empty set B such that Relation from A to B is a subset of $A \times B$ i.e. $R \subseteq A \times B$	• Function is a special type of relation from set A to set B where all the elements of set A are related to some or all elements of set B and no two elements of set B is related to a common element in set A .

• In case of relation of the pre-image may or may not have an image	• In case of function all pre-image must have an image.
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• A pre-image can have two or more image	• no-two image can have a common pre-image
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• A relation can or cannot be a function.	• A function is always a relation.
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Relation:

Definition - let A and B be two sets. A relation from A to B is a subset of $A \times B$.

Symbolically, R is a relation from A to B iff $R \subseteq A \times B$.

• A relation connecting two elements is called as binary relation and this can be expressed as a set whose elements are ordered pairs.

• In the ordered pair (a, b) if 'a is related to b' then we write aRb or $a^R b$.

Thus $(a, b) \in R \leftrightarrow aRb$
or $R = \{(a, b) \mid aRb\}$

• If in the ordered pair (a, b) 'a is not related to b' then we write ' $a \not R b$ '.

• Product Sets -

let A and B be any two sets. The cartesian product of A and B is denoted by $A \times B$, is the set of all ordered pairs of the form (a, b) where $a \in A$ and $b \in B$.

ie $A \times B = \{(a, b) \mid a \in A, b \in B\}$

eg let $A = \{a, b\}$ and $B = \{1, 2, 3\}$ Then

$$A \times B = \{a, b\} \times \{1, 2, 3\}$$

$$= \{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3)\}$$

• $A \times B \neq B \times A$

• Binary Relation -

A relation R between pairs of elements of a set A is called a binary relation.

The word relation will mean binary relation. Thus the sets of ordered pairs are binary relations.

In other words, A binary relation is a relation between two elements from two sets, say A & B . It is a subset of the cartesian product $A \times B$ which means it is a set of ordered pairs (a, b) where $a \in A$ and $b \in B$.

Formally, a binary relation R from A to B is defined as $R \subseteq A \times B$ where R is the binary relation.

A is the domain (set of inputs).

B is the co-domain (set of possible outputs).

$(a, b) \in R$ means a is related to b i.e. $a R b$.

eg. let $A = \{1, 2, 3\}$ and $B = \{a, b, c\}$

$R = \{(1, a), (2, b), (3, c)\}$ is a binary relation from A to B .

• Domain and Range of a Relation -

let R be a relation from A to B i.e. $R: A \rightarrow B$

Domain - The set of all first co-ordinates of the members of the relations set R is called the domain of R .

$$\text{Domain } R = \{x \mid (x, y) \in R\}$$

Range - The set of all second members of R is called the range of R .

$$\text{Range } R = \{y \mid (x, y) \in R\}$$

eg. $A = \{1, 2\}$, $B = \{a, b\}$

$$\therefore A \times B = \{(1, a), (1, b), (2, a), (2, b)\}$$

Domain $R = \{1, 2\}$ Range $R = \{a, b\}$

• Properties of Binary Relation -

Properties of relation R on a set A .

- Reflexive - A relation R in a set A is said to be reflexive if every element in A is related to itself.

ie if aRa for every $a \in A$

$$\text{ie } R = \{ (a, a) \mid a \in A \}$$

example,

$$A = \{ 1, 2, 3 \}$$

$R_1 = \{ (1, 1), (2, 2), (3, 3) \}$ is a reflexive relation

$R_2 = \{ (1, 1), (1, 2), (1, 3) \}$ is not reflexive relation

- Irreflexive - A relation R in a set A is said to be irreflexive if the element in A does not related to itself.

ie for all $a, a \in A, a \neq a$ ie $a \not R a$

- Symmetric - A relation R is said to be symmetric if $aRb \rightarrow bRa$ for all $a, b \in A$.

example,

$$A = \{ 1, 2, 3 \} \text{ Then the relations}$$

$$R_1 = \{ (1, 1), (2, 2), (3, 3) \} \text{ and } R_2 = \{ (1, 2), (2, 1), (1, 3), (3, 1) \}$$

are symmetric relations.

Mathematically,

if $a \in A, b \in A$ and $(a, b) \in R$ Then

$$\forall a, \forall b (aRb \rightarrow bRa)$$

- Asymmetric - If for all $a \in A$, and for all $b \in A$ if $(a, b) \in R$ then $(b, a) \notin R$.

Mathematically, $\forall a, \forall b [(aRb) \rightarrow \neg(bRa)]$.

example,

if A is set of all married peoples then if relation R is indicating husband hood then

aRb indicates that a is husband of $b \Rightarrow b \not R a$

ie b is not husband of a .

• Anti-symmetric relation -

An anti-symmetric relation is a binary relation R on a set A if
 $(a, b) \in R$ and $(b, a) \in R \rightarrow a = b$ for all $a, b \in R$

example, if $A = \{a, b, c, d\}$ then

$R = \{(a, c), (d, b), (c, d), (d, c)\}$ is not an anti-symmetric relation since $(c, d) \in R$ and $(d, c) \in R$ but $c \neq d$.

example, If $A \subseteq B$ and $B \subseteq A$, then $A = B$ is anti-symmetric.

- If $a \leq b$ and $b \leq a$ then $a = b$ is ex. of anti-sym.
- If $a \geq b$ and $b \geq a$ then $a = b$

• Transitive Relation -

A relation R on set A is said to be a transitive relation iff aRb and $bRc \Rightarrow aRc$ where $a, b, c \in A$.

Mathematically,

$$\forall a, \forall b, \forall c, [aRb \wedge bRc \rightarrow aRc]$$

• Non-transitive Relation -

If aRb and bRc then aRc for all $a, b, c \in A$

• Converse of relation -

If the given relation R is defined from A to B then the converse of relation, denoted by R^c is the relation from B to A defined as

$$R^c = \{(b, a) \mid (a, b) \in R\}$$

Clearly,

$$R^c \subseteq B \times A$$

Properties - Let R and S be relations from A to B then

$$(R^c)^c = R$$

$$(R \cup S)^c = R^c \cup S^c$$

$$(R \cap S)^c = R^c \cap S^c$$

Ex let $A = \{a, b, c, d\}$ and $B = \{1, 2, 3\}$

let $R = \{(a, 1), (c, 1), (c, 3)\}$

Find R^c , $D(R^c)$, $R_n(R^c)$

Soln Given $R = \{(a, 1), (c, 1), (c, 3)\}$

$$\therefore R^c = \{(1, a), (1, c), (3, c)\}$$

$$\therefore \text{Domain of } R^c = D(R^c) = \{1, 3\} = R_n(R)$$

$$\therefore \text{Range of } R^c = R_n(R^c) = \{a, c\} = D(R)$$

Ex let $A = \{1, 2, 3, 4\}$ let R and S be relations on A such that $R = \{(a, b) \mid a = b+1 \text{ or } b = 2a\}$ and $S = \{(a, b) \mid a \text{ divides } b\}$

Show that $(R \cap S)^c = R^c \cap S^c$

Soln $R = \{(2, 1), (3, 2), (4, 3), (1, 2), (2, 4)\}$ and

$$S = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}$$

$$\text{Then } R \cap S = \{(1, 2), (2, 4)\}$$

$$\therefore (R \cap S)^c = \{(2, 1), (4, 2)\} \dots (1) \text{ Also}$$

$$R^c = \{(1, 2), (2, 3), (3, 4), (2, 1), (4, 2)\}$$

$$\text{and } S^c = \{(1, 1), (2, 1), (3, 1), (4, 1), (2, 2), (4, 2), (3, 3), (4, 4)\}$$

$$\therefore (R^c \cap S^c) = \{(2, 1), (4, 2)\} \dots (2)$$

Hence from (1) and (2)

$$(R \cap S)^c = R^c \cap S^c$$

Representations of Relations on finite sets -

let A and B are finite sets.

The following are two ways of picturing a relation $R: A \rightarrow B$.

• Matrix of Relation -

If $A = \{a_1, a_2, \dots, a_m\}$ and $B = \{b_1, b_2, \dots, b_n\}$ are finite sets containing m and n elements respectively and R is relation from A to B , we represent R by the $m \times n$ matrix $M_R = [m_{ij}]$ defined as

$$m_{ij} = \begin{cases} 1 & \text{if } (a_i, b_j) \in R \\ 0 & \text{if } (a_i, b_j) \notin R \end{cases}$$

The matrix M_R is called matrix of relation R .
Method -

- Form a rectangular array whose rows are labelled by the elements of A and whose columns are labelled by the elements of B .
- Put 1 or 0 in each position of the array according as $(a, b) \in R$ or $(a, b) \notin R$ respectively.

Example -

let $A = \{1, 2, 3\}$ and $B = \{x, y, z\}$ and let $R = \{(1, y), (1, z), (3, y)\}$

$\therefore$ Matrix of relation $R: A \rightarrow B$ is given by

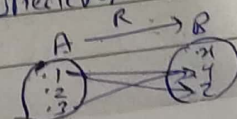
	x	y	z
1	0	1	1
2	0	0	0
3	0	1	0

$$\therefore M_R = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

• Arrow diagram of relation -

Method: Write the elements of A and B in two different disjoint discs and draw an arrow from $a \in A$ to $b \in B$ whenever aRb .

eg $R = \{(1, y), (1, z), (3, y)\}$



• Directed graphs of relations on sets -

(Diagram of a relation) :-

If A is a finite set and R is a relation on A .
Draw a small circle for each element of A and label the circle with the corresponding element of A . These circles are called vertices.

Draw an arrow called an edge, from the vertex a_i to vertex a_j if and only if $a_i R a_j$.

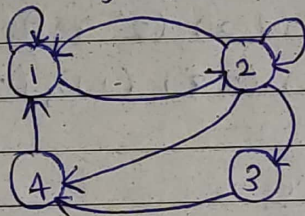
The resulting pictorial representation of R is called a directed graph or diagram of R .

Thus if R is relation on A , the edges in the diagram of R correspond exactly to the pairs in R and the vertices correspond exactly to the elements of a set A .

Example - let $A = \{1, 2, 3, 4\}$

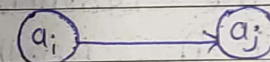
$R = \{(1,1), (1,2), (2,1), (2,2), (2,3), (2,4), (3,4), (4,1)\}$

Then the diagram of R is as shown below

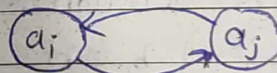


Note : If $(a_i, a_j) \in R$

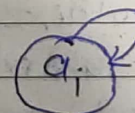
$(a_j, a_i) \in R$



$(a_i, a_j) \in R, (a_j, a_i) \in R$



$(a_i, a_i) \in R$



* Closures of Relation -

The closure of a relation R on a set A is defined as the smallest relation R' on A that contains R and satisfies certain properties.

Types of closure:

Reflexive closure :-

A reflexive closure of a relation R on a set A is the smallest reflexive relation R' that contains R .

A relation R is reflexive if $\forall a \in A, (a, a) \in R$

In other words, every element of A is related to itself.

The reflexive closure of R , denoted by $r(R)$ is defined as

$$r(R) = R \cup \{(a, a) \mid a \in A\}$$

This means that $r(R)$ contains all the elements of R , plus all the pairs (a, a) , $\forall a \in A$.

Example, let $A = \{1, 2, 3\}$ and

$$R = \{(1, 2), (2, 3)\}$$

$$\text{Then } r(R) = \{(1, 2), (2, 3), (1, 1), (2, 2), (3, 3)\}$$

Symmetric closure -

The symmetric closure of a relation R on a set A is the smallest symmetric relation R' that contains R .

A relation R is symmetric if :

$$\forall a, b \in A, (a, b) \in R \Rightarrow (b, a) \in R$$

In other words, if aRb then bRa .

The symmetric closure is denoted by $s(R)$ is defined as

$$s(R) = R \cup \{(b, a) \mid (a, b) \in R\}$$

This means that $s(R)$ contains all the elements of R plus all the pairs (b, a) $\forall$ pair $(a, b) \in R$.

eg, let $A = \{1, 2, 3\}$ and $R = \{(1, 2), (2, 3)\}$

$$\text{Then } s(R) = \{(1, 2), (2, 3), (2, 1), (3, 2)\}$$

• Transitive closure

The transitive closure of a relation R on set A is the smallest transitive relation R' that contains R .
A relation R is transitive if

$$\forall a, b, c \in A \quad (a, b) \in R \text{ and } (b, c) \in R \Rightarrow (a, c) \in R$$

The transitive closure of R is denoted by $t(R)$ is defined as

$$t(R) = R \cup \{ (a, c) \mid \exists b \in A, (a, b) \in R \text{ and } (b, c) \in R \}$$

This means that $t(R)$ contains all the elements of R plus all the pairs (a, c) for every pair (a, b) and (b, c) in R .

Example, let $A = \{1, 2, 3\}$ and $R = \{(1, 2), (2, 3)\}$

$$\text{Then } t(R) = \{(1, 2), (2, 3), (1, 3)\}$$

Equivalence Relation

A relation R on a set A is said to be an equivalence relation if it is reflexive, symmetric and transitive.

In other words,

An equivalence relation is a binary relation R on a set A that satisfies three properties:

- 1) Reflexivity: For all $a \in A$, $(a, a) \in R$.
- 2) Symmetric: For all $a, b \in A$, $(a, b) \in R \Rightarrow (b, a) \in R$.
- 3) Transitivity: For all $a, b, c \in A$,
 $(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$.

Ex Let T be a set of triangles in a plane and define R as the set $R = \{(a, b) \mid a, b \in T, a \text{ is congruent to } b\}$
Then show that R is an equivalence relation.

Soln If a and b are triangles in a plane then $(a, b) \in R$ iff a is congruent to b .

(Two Δ are congruent if the area of one triangle is equal to the area of the other)

$$\text{ie } \Delta a = \Delta b$$

Reflexive -

Since $\Delta a = \Delta a \quad \forall a \in T$

a is congruent to a , $(a, a) \in R$

Hence R is reflexive

Symmetry -

$(a, b) \in R \Rightarrow a$ is congruent to b

$$\Rightarrow \Delta a = \Delta b$$

$\Rightarrow \Delta b = \Delta a \Rightarrow b$ is congruent to a

$$\Rightarrow (b, a) \in R$$

Hence R is symmetric

For Transitivity -

$(a, b) \in R, (b, c) \in R$

$\Rightarrow a$ is congruent to b and b is congruent to c

$$\Rightarrow \Delta a = \Delta b \text{ and } \Delta b = \Delta c$$

$$\Rightarrow \Delta a = \Delta c$$

$\Rightarrow a$ is congruent to c

$$\Rightarrow (a, c) \in R$$

Hence R is transitive

Thus R is an equivalence relation.

* Partial Ordering Relations -

A relation R on a set A is called a partial ordering of a partial order if R is reflexive, antisymmetric and transitive.

A set A together with a partial ordering R i.e. $(A, \leq)$ is called a partially ordered set or poset.

Furthermore, if a relation R in A defines a partial order then $(a, b) \in R$ denoted by $a \leq b$ read as ' a precedes b '.

In other words,

A partial ordering relation also known as a partial order, is a binary relation $\leq$ on a set P that satisfies the following properties:

a) Reflexivity - For all $a \in P$, $a \leq a$

2) Antisymmetry - For all $a, b \in P$,
if $a \leq b$ and $b \leq a$ then $a = b$

3) Transitivity - For all $a, b, c \in P$
if $a \leq b$ and $b \leq c$ then $a \leq c$.

Examples:

- The relation $\subseteq$ of set inclusion is a partial ordering on any collection sets since set inclusion has three desired properties i.e.

i) $A \subseteq A$ for any set A (Reflexive)

ii) If $A \subseteq B$ and $B \subseteq A \Rightarrow A = B$ (Anti-symmetry)

iii) If $A \subseteq B$ and $B \subseteq C \Rightarrow A \subseteq C$ (Transitive)

- The relation $\leq$ on the set R of real numbers is reflexive, anti-symmetric and Transitive.

Thus $\leq$ is a partial ordering.

Partition -

- Let A be any non-empty set. A partition of A is a set of non-empty subsets of A denoted $\{A_1, A_2, \dots, A_k\}$ such that their union is equal to A and the intersection of A_i and A_j is empty for any distinct A_i and A_j .

- In other words, a partition of a set is a division of the elements in the set into disjoint subsets. These subsets are called blocks of the partition.

Ex. Let $A = \{a, b, c, d, e, f, g\}$. Then $\{\{a\}, \{b, c, d\}, \{e, f\}, \{g\}\}$ is a partition of A .

Ex. Find all possible partitions of $A = \{a, b, c, d\}$

Solution We have fifteen partitions for given set A .

- $\{\{a, b, c, d\}\}$
- $\{\{a\}, \{b, c, d\}\}, \{\{b\}, \{a, c, d\}\}, \{\{c\}, \{a, b, d\}\}, \{\{d\}, \{a, b, c\}\}$
- $\{\{a, b\}, \{c, d\}\}, \{\{a, c\}, \{b, d\}\}, \{\{a, d\}, \{b, c\}\}$

$\{\{a\}, \{b\}, \{c, d\}\}, \{\{a\}, \{c\}, \{b, d\}\},$
 $\{\{a\}, \{d\}, \{b, c\}\}, \{\{b\}, \{c\}, \{a, d\}\},$
 $\{\{b\}, \{d\}, \{a, c\}\}, \{\{c, d\}, \{a, b\}\},$
 $\{\{a\}, \{b\}, \{c\}, \{d\}\}.$

Rules to find the partitions of a set -

Singletons - Each element can be a separate partition.

Pairs - Choose 2 elements to form a pair, and the remaining elements can be singletons or form another pair.

Triples - Choose 3 elements to form a triplet, and the remaining elements can be singletons or form a pair.

Quadruplets - Choose 4 elements to form quadruplets (only possible if the set has 4 elements).

Combine - Combine the above rules to form large partitions.

Disjoint - Ensure that the partitions are disjoint (no element is in more than one partition).

Exhaustive - Ensure that every element is in exactly one partition.

Non empty - Ensure that each partition is non-empty (has at least one element).

Hasse Diagram

A Hasse diagram is a graphical representation of a partially ordered set (poset).

It is useful for visualizing the relationships between elements in a poset.

Rules to draw Hasse diagram -

1) Start with the minimal element:

Begin by placing the minimal elements (ie those with no predecessors) at the bottom of diagram.

2) Draw nodes:

Represent each element in the poset as a node (circle or point).

3) Draw edges:

Connect nodes with edges to represent the partial order relation ($\sim$) (read as $\tilde{}$ tilde).

If $a \sim b$, draw an edge from a to b .

4) Direction matters -

Edges are directed, pointing from the smaller elt to the larger element.

5) Transitive reduction -

Omit edges that can be formed by transitivity.

If $a \sim b$ and $b \sim c$ then don't draw edge from a to c .

6) No loops -

Don't draw loops, as the relation is reflexive ($a \sim a$).

7) No bidirectional edges -

Don't draw bidirectional edges as the relation is antisymmetric (if $a \sim b$ and $b \sim a$ then $a = b$).

8) Minimize crossings -

Arrange nodes to minimize edge crossing for clarity.

9) Label nodes -

Label each node with the corresponding element.

10) Check for consistency -

Verify that the diagram accurately represents the partial order relation.

Ex. Let $A = \{1, 2, 3, 4, 6\}$ and R be the relation
 aRb iff a is divisor of b .

Draw the Hasse diagram.

Soln Given: $A = \{1, 2, 3, 4, 6\}$

Since R defines as ' a is divisor of b '

$$R = \{(1,1), (1,2), (1,3), (1,4), (1,6), \\ (2,4), (2,6), (3,6), (4,4)\}$$

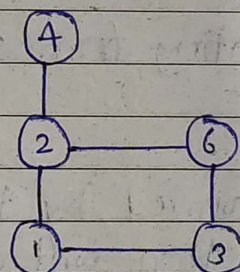
Here, 1 is the minimal element i.e. 1 has no any predecessor. So we draw a node (circle or dot) to represent 1 at bottom of diagram.

Now elements 2, 3, 4 and 6 are connected to 1 also 4 and 6 are connected to 2 and element 6 is connected to 3.

i.e. 2 and 3 has predecessor 1.

6 has predecessor 2 and 3.

4 has predecessor 2.



Hasse diagram.

Ex. If $A = \{1, 2, 3, 4\}$

$$R = \{(1,1), (1,2), (2,2), (2,4), (1,3), (3,3), (3,4), (1,4), \\ (4,4)\}$$

Then show that R is a partial order and draw its Hasse diagram.

Soln To show R is partial order, we show R is reflexive, antisymmetric and transitive.

R is reflexive since $(1,1), (2,2), (3,3), (4,4) \in R$.

R is anti-symmetric since

$$(1,2) \in R \text{ but } (2,1) \notin R, (2,4) \in R \text{ but } (4,2) \notin R$$

$$\text{Similarly } (1,3), (1,4), (3,4) \in R \text{ but } (3,1), (4,1), (4,3) \notin R$$

Also R is transitive

as $(1, 2) \in R, (2, 4) \in R \Rightarrow (1, 4) \in R$

$\therefore R$ is a partial order relation on set A .

Now

To draw Hasse diagram, we have

$R = \{(1, 1), (1, 2), (2, 2), (2, 4), (1, 3), (3, 3), (3, 4), (1, 4), (4, 4)\}$

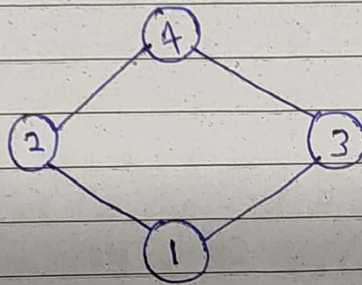
neglate / delete pair of the form $(a, a) \forall a \in A$.

$\therefore R = \{(1, 2), (2, 4), (1, 3), (3, 4), (1, 4)\}$

As 1 connect to 2, 2 connect to 4

1 connect to 3, 3 connect to 4

So, neglate edge from 1 to 4.



• 1 has no predecessor so

1 has node at bottom

• 2 and 3 has predecessor

1

• 4 has predecessor 2 and 3



Lattice -

It is a partial ordered set in which every two elements have a unique upper bound and a unique greatest lower bound.

In other words,

A poset $(L, \leq)$ is called a lattice if every two element subset of L has a unique LUB and GLB. It is denoted by $[L, \vee, \wedge]$

• Greatest element (Maximal element) -

An element $a \in A$ is called a maximal element if there is no element $b \in A$ such that $b \neq a$ and $a \leq b$.

• Least element (Minimal element) -

An element $c \in A$ is called a minimal element if there is no element $d \in A$ such that $d \neq c$ and $d \leq c$.

Example,

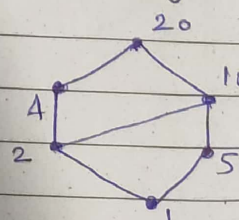
Let D_n be set of all divisor of n where n is a positive integer and consider the relation divides, then $(D_n, \leq)$ is a poset.

$(D_{20}, \leq)$ is a poset where $\leq$ is divider.
 $D_{20} = \{1, 2, 4, 5, 10, 20\}$

Relation set $\leq = \{(1,1), (1,2), (1,4), (1,5), (1,10), (1,20), (2,2), (2,4), (2,10), (2,20), (4,4), (4,20), (5,5), (5,10), (5,20), (10,10), (10,20), (20,20)\}$

Maximal element = 20

Minimal element = 1



Hasse diagram

• Lower bound -

let $(L, \leq)$ be a poset, $a, b \in L$ then
 $c \in L$ is a lower bound of a and b if
 $c \leq a$ and $c \leq b$.

• Upper bound -

let $(L, \leq)$ be a poset, $a, b \in L$ then $d \in L$ is
 a upper bound of a and b if $a \leq d$ and $b \leq d$.

• Greatest lower bound (GLB) -

let $(L, \leq)$ be a poset. @
~~a~~ $a, b \in L$

An element $g \in L$ is said to be a greatest lower
 bound (GLB) or infimum of a and b if

- i) g is a lower bound of a and b
- ii) there is no other lower bound g' of a and b
 such that $g \leq g'$.

• Least Upper bound (LUB) -

let $(L, \leq)$ be a poset, $a, b \in L$.

An element $I \in L$ is said to be a least upper
 bound [LUB] or supremum of a and b if

- i) I is the upper bound of a and b .
- ii) There is no other upper bound I' of a and b
 such that $I' \leq I$.

• Join and Meet -

let $(L, \leq)$ be a poset, $a, b \in L$

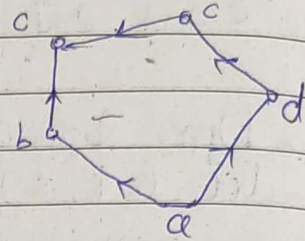
$$a \vee b = \text{lub}(a, b) \quad (\vee = \text{join})$$

$$a \wedge b = \text{GLB}(a, b) \quad (\wedge = \text{meet})$$

eg $[P(A), \vee, \wedge]$ is a lattice under the relation sub
 where $A = \{a, b, c\}$

• Chain -

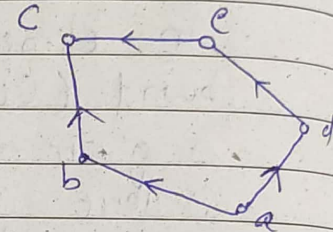
Let $(A, \leq)$ be a poset. A subset of A is called a chain if every two elements in the subset are related.



For the partially ordered relation

$\{a, b, c, e\}$, $\{a, b, c\}$, $\{a, d, e\}$, $\{a\}$ are chains.

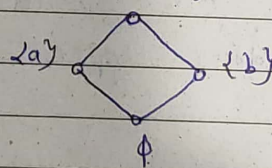
If A itself is a chain the poset $(A, \leq)$ is called a totally ordered set or linearly ordered set.



• Anti-chain -

A subset of A is called an anti-chain if not two distinct elements in the subset are related.

eg. let $A = \{a, b\}$ and consider its poset $(P(A), \subseteq)$



Following subsets are chain:

$\{\phi, \{a\}, \{a, b\}\}$, $\{\phi, \{b\}, \{a, b\}\}$, $\{\{a\}, \{a, b\}\}$, $\{\{b\}, \{a, b\}\}$, $\{\phi, \{a\}\}$, $\{\phi, \{b\}\}$

The subset $\{\{a\}, \{b\}\}$ is an anti-chain.

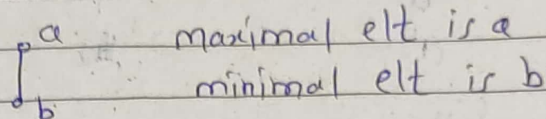
• Lattice -

It is a partial ordered set in which every two elements have a unique upper bound and a unique greatest lower bound.

• Maximal and Minimal elements -

An element $a \in A$ is called a maximal element if there is no element $b \in A$ such that $b \neq a$ and $a \leq b$.

An element $c \in A$ is called a minimal element if there is no element $d \in A$ such that $d \neq c$ and $d \leq c$.



• Upper bound and lower bound -

Let a and b be two elements in a partially ordered set $(A, \leq)$.

An element c is said to be an upper bound of a and b if $a \leq c$ and $b \leq c$.

An element c is said to be least upper bound (l.u.b.)

A element c is said to be lower bound of a and b if $c \leq a$ and $c \leq b$ and an element c is said to be a greatest lower bound (i.e. g.l.b.) of a and b if c is a lower bound of a and b and if there is no other lower bound d of a and b such that $c \leq d$.

- least upper bound (l.u.b.) is known as supremum
- greatest lower bound (g.l.b.) is known as infimum

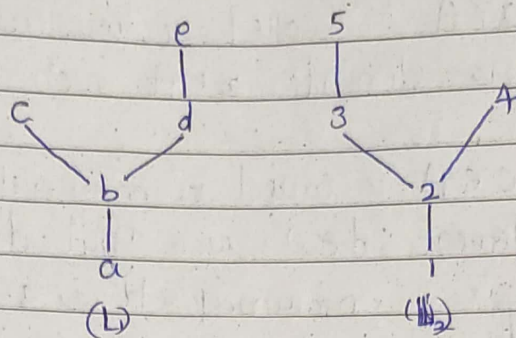
• Isomorphic Lattices -

Two lattices L_1 and L_2 are said to be isomorphic if there is a function $f: L_1 \rightarrow L_2$ such that

- f is one-one
- f is onto
- $f(a \wedge b) = f(a) \wedge f(b)$
- $f(a \vee b) = f(a) \vee f(b)$

Note: l.u.b. of a and b is $a \vee b$ called join of a and b .
g.l.b. of a and b is $a \wedge b$ called meet of a and b .

Ex Are the two lattices are isomorphic?



Soln

For lattices L_1 and L_2 , we defined a mapping $f: L_1 \rightarrow L_2$

1) one-one

$$f(a) = 1, f(b) = 2, f(c) = 4, f(d) = 3, f(e) = 5$$

$\therefore f$ is one-one mapping

2) onto

$$1 = f(a), 2 = f(b), 3 = f(d), 4 = f(c), 5 = f(e)$$

$\therefore f$ is onto mapping

3) let for $\{c, d\} \in L_1$,

$$f(c \vee d) = f(c) \vee f(d)$$

$$f(e) = 4 \vee 3$$

$$5 = 5$$

$$\therefore f(c \vee d) = f(c) \vee f(d)$$

$$4) f(c \wedge d) = f(c) \wedge f(d)$$

$$f(b) = 4 \wedge 3$$

$$2 = 2$$

$$\therefore f(c \wedge d) = f(c) \wedge f(d)$$

Here all four cases of isomorphism of lattices are verified or satisfied.

$\therefore$ lattices L_1 and L_2 are isomorphic

- Warshall's Algorithm: (To find transitive closure of a relation) -

Ex let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 2), (2, 3), (3, 4), (2, 1)\}$
find the transitive closure of R by using Warshall's algorithm.

Soln Given $A = \{1, 2, 3, 4\}$

here $|A| = 4$

So, Algorithm contains 5 matrices w_0, w_1, w_2, w_3, w_4
where $w_0 = M_R$

Given $R = \{(1, 2), (2, 3), (3, 4), (2, 1)\}$

The matrix corresponding to relation R is

$$w_0 = M_R = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

To find w_1 : consider the non-zero entries in first column and first row of the matrix w_0 .

$$C_1: \{2\}$$

$$R_1: \{2\}$$

$$C_1 \times R_1 = \{(2, 2)\}$$

$\therefore$ New entry in w_1 is $(2, 2)$, remaining all entries in w_1 same as w_0 .

$$\therefore w_1 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

For w_2 :

$$C_2: \{1, 2\}$$

$$R_2: \{1, 2, 3\}$$

$$C_2 \times R_2 = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3)\}$$

$$\therefore w_2 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

For W_3 :

$$C_3 = \{1, 2\}$$

$$R_3 = \{4\}$$

$$C_3 \times R_3 = \{(1, 4), (2, 4)\}$$

$$\therefore W_3 = \begin{matrix} & 1 & 2 & 3 & 4 \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

For W_4 :

$$C_4 = \{1, 2, 3\}$$

$$R_4 = \emptyset$$

$$\therefore C_4 \times R_4 = \{1, 2, 3\} \times \emptyset$$

= no ordered pairs

$\therefore W_4$ same as W_3 i.e. $W_4 = W_3$

$$\therefore W_4 = \begin{matrix} & 1 & 2 & 3 & 4 \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$\therefore R^* = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 1), (2, 2), (2, 3), (2, 4), (3, 4)\}$$



• Function -

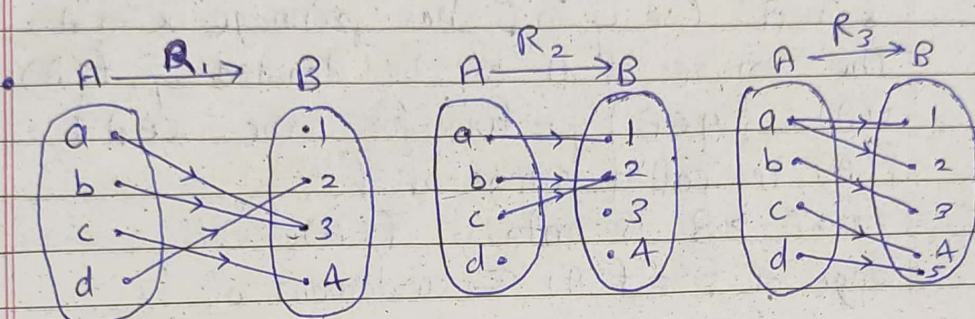
Definition - A function (or mapping) f from a set A to set B ($f: A \rightarrow B$) is a relation which associates for each element x in A , a unique (exactly one) element y in B .

Then the element y is expressed as $y = f(x)$.

y is the image of x under f .

f is also called a map or transformation.

• $f: A \rightarrow B$, here A is called domain of f and B is called the co-domain of f .



In $R_1: A \rightarrow B$

Since every element from A is associated to exactly one element in B .

So R_1 is a well defined function.

In $R_2: A \rightarrow B$

Relation R_2 is not a function because element d in A is not associated to any element in B .

In $R_3: A \rightarrow B$

Relation R_3 is again not function because element a in A is associated to two elements in B .

ie $R_1(a) = 1$, $R_3(a) = 2$.

• Types of function -

One-one or One to one or Injective function -

Definition - A function $f: A \rightarrow B$ is said to be one-one if different elements in A have different images in B .

$$f(a) = f(b) \Rightarrow a = b \quad [As \ a \neq b \Rightarrow f(a) \neq f(b)]$$

Onto or Surjective function -

Definition - A function $f: A \rightarrow B$ is said to be onto if every element y in B is an image of some x in A (or y in B has preimage x in A).

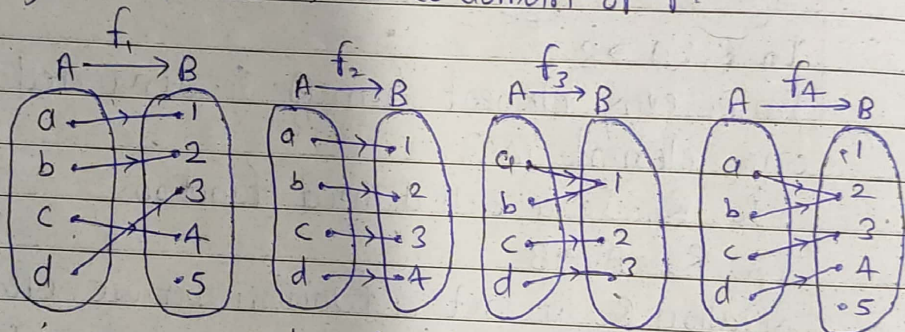
The image of A can be denoted by $f(A)$

$$f(A) = \{y \in B \mid y = f(x) \text{ for some } x \in A\}$$

$f(A)$ is called range of f .

• $f: A \rightarrow B$ is onto if $f(A) = B$

range of $f = f(A) \subset \text{co-domain of } f$.



f_1 is one-one but not onto as elt 5 is in B has no pre-image in A .

f_2 is one-one and onto

f_3 is onto but not one-one as $f(a) = f(b) = 1$ but at

f_4 is neither one-one nor onto.

Constant function -

$$f(x) = k \quad k \in \mathbb{R}$$

eg. $f(x) = 2$

Domain: $\mathbb{R}$ or $(-\infty, \infty)$

Range: $\{k\}$

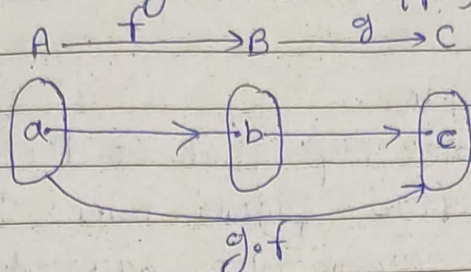
Identity function -

If $f: \mathbb{R} \rightarrow \mathbb{R}$ then identity function is defined by $f(x) = x$ for every $x \in \mathbb{R}$.

• Composition of function -

Let A, B, C be three sets and f be a mapping from A to B . i.e. $f: A \rightarrow B$.

And let g be a mapping from B to C i.e. $g: B \rightarrow C$.



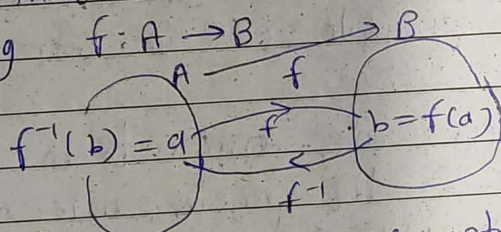
Then $g \circ f$ composite of f and g is a function from A to C .

$$\begin{aligned} g \circ f(a) &= g[f(a)] \\ &= g[b] \\ &= c \quad \forall a \in A. \end{aligned}$$

Domain of $g \circ f$ is set A and co-domain of $g \circ f$ is set C .

• Inverse function -

If $f: A \rightarrow B$ is an one-one, onto (bijective) map then the mapping $f^{-1}: B \rightarrow A$ which associates to element $b \in B$ to the unique element $a \in A$ such that $f(a) = b$ is called the inverse of mapping $f: A \rightarrow B$.

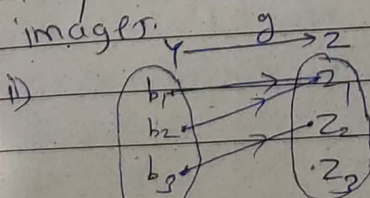


If $f(a) = b$ then
 $f^{-1}(b) = a$.

1) If $f: A \rightarrow B$ is one-one, onto mapping then only f^{-1} exist and f^{-1} is also one-one onto mapping.

2) In case of one one into, many one into and many one into inverse does not exist as in case a).

1) For $f^{-1}: B \rightarrow A$, $B \in B$ is not corresponding with any element of A and Z is having two different images.



hence f^{-1} does not exist.

iii) for h^{-1} , $B_1 \rightarrow A_1$, $b_3 \in B$ is having two different images hence h^{-1} does not exist.
Hence f^{-1} exists only in case f is one-one and onto mapping.

• Pigeonhole principle -

Pigeonhole principle says that if there are many pigeons and a few pigeon holes, then there must be some pigeonholes occupied by two or more pigeons.

Let A and B be finite sets and $|A| > |B|$ then for any function f from A to B , $\exists a_1, a_2 \in A$ such that $f(a_1) = f(a_2)$.

• The extended Pigeonhole principle -

If there are m pigeon holes and more than $3m$ pigeons then 4 or more pigeons will occupy at least one of the pigeon holes.

In general if the number of pigeons is much larger than the number of pigeon holes, the pigeonhole principle can be extended as -

Theorem -

If n pigeons are assigned to m pigeon holes, then one of the pigeon holes must be occupied by at least $\left\lceil \frac{n-1}{m} \right\rceil + 1$ pigeons.

Recursive Function -

We have two steps:

Basic Step - Specify the value of the function at zero.

Recursive Step - Give a rule to define the function with the set of non-negative integers.

Note - Domain must be well defined.

Domain - set of non-negative integers

Example -

Basic step - $f(0) = 2$

Recursive step - $f(n+1) = 3f(n) + 4$

Find $f(1)$, $f(2)$, $f(3)$, $f(4)$.

Solution -

Given: $f(0) = 2$ } given in question
 $f(n+1) = 3f(n) + 4$

For $n=0$, $f(0+1) = 3f(0) + 4$
 $= 3(2) + 4$

$$\boxed{f(1) = 10}$$

$n=1$, $f(1+1) = 3f(1) + 4$
 $= 3(10) + 4$

$$\boxed{f(2) = 34}$$

$n=2$, $f(2+1) = 3f(2) + 4$
 $= 3(34) + 4$

$$\boxed{f(3) = 106}$$

$n=3$, $f(3+1) = 3f(3) + 4$
 $= 3(106) + 4$

$$\boxed{f(4) = 322}$$

- How to write a Recursive formula for Arithmetic sequence?

Steps to find the recursive formula for the arithmetic sequence -

Step 1 - Determine whether the given sequence is arithmetic. (Add or subtract two successive terms. If we get the same amount from one term to the next term, then the sequence is arithmetic.)

Step 2: Find the common difference of the given sequence.

Step 3: Formulate the recursive formula by stating the first term and then create the formula to be the previous term + common difference. Thus, the recursive formula for the Arithmetic sequence is given as:

$$a_n = a_{n-1} + d$$

eg. Consider the sequence: 2, 4, 6, 8, 10, ...

The above sequence is an Arithmetic sequence because each term in the sequence is increased by 2.

Hence the common difference is 2.

Thus, the recursive formula is

$$a_n = a_{n-1} + 2$$

Explanation:

here, $a_1 = 2$

$$a_2 = a_{(2-1)} + 2$$

$$= a_1 + 2$$

$$= 2 + 2$$

$$a_2 = 4$$

$$a_3 = a_{(3-1)} + 2$$

$$= a_2 + 2$$

$$= 4 + 2$$

$$a_3 = 6$$

$$a_4 = a_{4-1} + 2 = a_3 + 2 = 6 + 2 = 8$$

and the process continues.

- How to write a Recursive formula for a geometric sequence?

Steps to find the recursive formula for geometric sequence -

Step 1 - Determine whether the given sequence is geometric. (Multiply or divide each term by a number. If we get the same amount from one term to the next term, then the sequence is geometric.)

Step 2: Find the common ratio of the given sequence.

Step 3: Formulate the recursive formula by stating the first term, and then create the formula to be the common ratio to the previous term.

Thus the recursive formula for the geometric sequence is given as:

$$a_n = r \cdot a_{n-1}$$

eg. Consider the sequence: 3, 15, 75, 375, ...

The above sequence is a geometric sequence because the successive term in the sequence is obtained by multiplying 5 to the previous term.

Hence the common ratio $r = 5$.

Thus the recursive formula for the sequence is

$$a_n = 5 \cdot a_{n-1}$$

Explanation -

Here $a_1 = 3$

$$\therefore a_2 = a_{(2-1)} \cdot 5 = a_1 \cdot 5 = 3(5) = 15$$

$$a_3 = a_2 \cdot 5 = 15 \cdot 5 = 75$$

$$a_4 = a_3 \cdot 5 = 75 \cdot 5 = 375$$

and the process continues.